

ENTERPRISE AWS CLOUD MIGRATION PLAYBOOK —

CHAPTER 1

Executive Strategy: 6R Migration Framework

Our enterprise cloud transition adopts the AWS 6R framework: Rehost, Replatform, Repurchase, Refactor, Retire, and Retain. Phase 1 targets the complete migration of 42 core microservices to Amazon EKS within 180 days.

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CHAPTER 2

Networking Architecture & VPC Peering

The multi-account structure features dedicated Ingress, Egress, Shared Services, and Workload VPCs connected via AWS Transit Gateway. Transit Gateway encryption uses AWS Key Management Service (KMS) customer-managed keys.

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CHAPTER 3

Kubernetes EKS Cluster Sizing & Node Pools

Production clusters utilize Karpenter for dynamic node provisioning. The base pool runs 12 m6i.4xlarge EC2 instances spanning 3 Availability Zones (us-east-1a, 1b, 1c) with pod anti-affinity guarantees.

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CHAPTER 4

Database Migration via AWS DMS

PostgreSQL 14 databases (totaling 8.4TB) are synchronized to Amazon Aurora PostgreSQL using AWS Database Migration Service (DMS) with continuous Change Data Capture (CDC). Target cutover downtime window: < 15 minutes.

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CHAPTER 5

Security Compliance, IAM Roles & KMS

IAM Roles for Service Accounts (IRSA) replace static AWS credentials in containers. All S3 buckets enforce Object Lock in compliance mode and default SSE-KMS encryption.

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CHAPTER 6

Disaster Recovery RTO and RPO Guarantees

Multi-region active-passive failover between us-east-1 and us-west-2 ensures a Recovery Time Objective (RTO) under 30 minutes and a Recovery Point Objective (RPO) under 5 minutes for tier-1 workloads.

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CHAPTER 7

Cost Optimization & Spot Instance Policies

Non-production test environments run exclusively on EC2 Spot Instances, reducing compute costs by 68%. All unused resources are automatically scaled to zero outside business hours using Kubernetes Keda cron triggers.